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COURSE CODE: SBT-DF203
LAB NAME: DNS INTRODUCTION AND TRAFFIC ANALYSIS
DATE: 21/09/2026
INSTRUCTOR: AMINU IDRIS

**Important Pre-Lab
Instruction**

Read the matching lecture slide deck before executing the practical. Work only inside the ICdfa-approved lab or your own isolated virtual environment. Record every command, observation and screenshot as contemporaneous evidence.

Course Code	SBT-DF203
Course Title	Basic Networking Skills for Digital Forensics
Lab Number	Lab 7
Lab Title	DNS Introduction and Traffic Analysis
Delivery Block	3/3 of 3
Scheduled Dates	19-25 September 2026
Estimated Duration	3-4 hours practical work plus report writing
Mode	Individual practical lab in an authorized virtual environment
Required Evidence	dig_dns.pcap or locally generated DNS captures; optional smtp.pcap correlation

1. Source Materials and Slide Alignment

- Domain Name System Forensics: domain-to-IP resolution, hierarchical DNS, resolver/TLD/authoritative roles, dig queries, capture with tshark, query/response fields and DNS within email traffic.

- The slide practical captures dig and browser DNS traffic, examines A records, local resolver configuration, TTL and authoritative information.
- The exercise builds the normal DNS baseline required before analyzing spoofing in Lab 8.

Reading Requirement Before Lab Execution

The lab deliberately follows the sequence, terminology and core exercises presented in the uploaded SBT-DF203 slides. Commands have been corrected where needed for Linux syntax and forensic repeatability.

2. Lab Scenario

You are documenting normal DNS behaviour for an incident-response baseline. You will identify the configured resolver, capture controlled dig queries, analyze transaction IDs, names, record types, response codes, answers and TTL values, and correlate DNS resolution with a subsequent web or email connection.

3. Learning Outcomes

- Explain recursive and iterative DNS resolution at a foundational level.
- Use dig to query A, AAAA, MX and NS records.
- Capture and analyze DNS queries and responses with tshark/Wireshark.
- Identify transaction ID, query name, record type, response code, answer and TTL.
- Correlate DNS responses with later IP connections.
- Recognize normal variations such as multiple answers, CNAMEs, cache responses and IPv6 queries.

4. Legal, Ethical and Safety Rules

- Use only systems, packet captures and networks for which you have explicit authorization.
- Keep the lab isolated from production, campus, office and public networks. Use NAT or host-only virtual networking as instructed.
- Do not collect, disclose or reuse real credentials, personal messages or confidential traffic.
- Preserve original capture files. Perform analysis on verified working copies and record SHA-256 hashes.
- Stop any traffic-generation or interception process immediately after the required evidence is captured.
- Restore firewall, ARP, forwarding and network settings before closing the lab.

5. Required Tools and Environment

Item	Recommended Tool	Purpose
Operating system	Kali Linux or Ubuntu VM	Generate and analyze DNS traffic.
DNS utility	dig from dnsutils	Perform controlled queries.

Item	Recommended Tool	Purpose
Capture tools	tshark and Wireshark	Capture DNS traffic.
Evidence	dig_dns.pcap or new PCAPNG	Analysis source.
Resolver config	/etc/resolv.conf, resolvectl	Identify configured DNS service.
Optional browser	Firefox/Chrome	Generate multiple DNS lookups.

6. Lab Folder Structure and Evidence Preparation

```
mkdir -p ~/SBT-DF203-Lab7/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab7
pwd
find . -maxdepth 1 -type d -print
```

Create the folder structure before downloading or generating evidence. Store original captures under evidence and analysis copies under working.

```
sudo apt update
sudo apt install -y dnsutils tshark wireshark
cat /etc/resolv.conf | tee reports/resolv_conf.txt
resolvectl status 2>/dev/null | tee reports/resolvectl_status.txt || true
wget -O evidence/dig_dns.pcap \
'https://raw.githubusercontent.com/frankwxu/digital-forensics-
lab/main/Networking_Forensics/lab_files/dns/dig_dns.pcap'
cp --preserve=timestamps evidence/dig_dns.pcap working/dig_dns_working.pcap
sha256sum evidence/dig_dns.pcap working/dig_dns_working.pcap | tee reports/dns_capture_hashes.txt
```

Screenshot required: Configured resolver information, evidence details and hashes.

ANSWER:

```
Kali Linux - VMware Workstation
File Edit View VM Tabs Help

Library
Type here to search
My Computer
  Owasip bvwa
  Metasploitable 2
  Kali Linux

Suspend this guest operating system
collins@kali: ~/SBT-DF203-Lab7

(collins@kali) - [~/Desktop]
$ mkdir -p ~/SBT-DF203-Lab7/{evidence,working,exported,reports,screenshots,scripts}

(collins@kali) - [~/Desktop]
$ cd ~/SBT-DF203-Lab7

(collins@kali) - [~/SBT-DF203-Lab7]
$ pwd
/home/collins/SBT-DF203-Lab7

(collins@kali) - [~/SBT-DF203-Lab7]
$ find . -maxdepth 1 -type d -print
.
./evidence
./screenshots
./scripts
./exported
./reports
./working

(collins@kali) - [~/SBT-DF203-Lab7]
$ sudo apt update
[sudo] password for collins:
Get:1 https://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 https://kali.download/kali kali-rolling/main amd64 Packages [21.6 MB]
Get:3 https://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.9 MB]
Get:4 https://kali.download/kali kali-rolling/contrib amd64 Packages [116 kB]
Get:5 https://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [270 kB]
Fetched 75.9 MB in 9min 7s (139 kB/s)
1307 packages can be upgraded. Run 'apt list --upgradable' to see them.
Warning: Problem unlinking the file prtial - pkgAcquire::Clean (21: Is a directory)
```

```
(collins@kali) ~/SBT-DF203-Lab7
$ sudo apt install -y dnsutils tshark wireshark
Note, selecting 'bind9-dnsutils' instead of 'dnsutils'
tshark is already the newest version (4.6.6-1).
wireshark is already the newest version (4.6.6-1).
The following packages were automatically installed and are no longer required:
  libpython3.13-dev python3.13 python3.13-dev python3.13-minimal python3.13-venv rpcsvc-proto
Use 'sudo apt autoremove' to remove them.
```

```
Upgrading:
  bind9-dnsutils bind9-host bind9-libs
```

```
Summary:
Upgrading: 3, Installing: 0, Removing: 0, Not Upgrading: 1304
Download size: 1,534 kB
Space needed: 188 kB / 59.0 GB available
```

```
Get:1 https://kali.download/kali kali-rolling/main amd64 bind9-dnsutils amd64 1:9.20.27-2 [164 kB]
Get:2 https://kali.download/kali kali-rolling/main amd64 bind9-host amd64 1:9.20.27-2 [52.2 kB]
Get:3 https://kali.download/kali kali-rolling/main amd64 bind9-libs amd64 1:9.20.27-2 [1,317 kB]
Fetched 1,534 kB in 12s (126 kB/s)
(Reading database... 426130 files and directories currently installed.)
Preparing to unpack ./bind9-dnsutils_1%3a9.20.27-2_amd64.deb...
Unpacking bind9-dnsutils (1:9.20.27-2) over (1:9.20.23-1)...
Preparing to unpack ./bind9-host_1%3a9.20.27-2_amd64.deb...
Unpacking bind9-host (1:9.20.27-2) over (1:9.20.23-1)...
Preparing to unpack ./bind9-libs_1%3a9.20.27-2_amd64.deb...
Unpacking bind9-libs:amd64 (1:9.20.27-2) over (1:9.20.23-1)...
Setting up bind9-libs:amd64 (1:9.20.27-2)...
Setting up bind9-host (1:9.20.27-2)...
Setting up bind9-dnsutils (1:9.20.27-2)...
Processing triggers for libc-bin (2.43-4)...
Processing triggers for man-db (2.13.1-1)...
Processing triggers for kali-menu (2026.2.6)...
```

```
(collins@kali) ~/SBT-DF203-Lab7
$ cat /etc/resolv.conf | tee reports/resolv_conf.txt
resolvectl status >/dev/null | tee reports/resolvectl_status.txt || true
# Generated by NetworkManager
search localdomain
nameserver 192.168.188.2
Do you want to install it? (N/y)y
Reading package lists...
Building dependency tree...
Reading state information...
Solving dependencies...
The following packages were automatically installed and are no longer required:
  libpython3.13-dev python3.13-dev python3.13-venv
  python3.13 python3.13-minimal rpcsvc-proto
Use 'sudo apt autoremove' to remove them.
```

```
Upgrading:
  libnss-systemd libsystemd0 systemd-sysv udev
  libpam-systemd libudev1 systemd-timesyncd
  libsystemd-shared systemd systemd-userdbd
```

```
Installing:
  systemd-resolved
```

```
Installing dependencies:
  libtss2-rc0t64 systemd-tpm
```

```
Suggested packages:
  libnss-myhostname libnss-resolve
```

```
Recommended packages:
  tpm-udev
```

```
Summary:
Upgrading: 10, Installing: 3, Removing: 0, Not Upgrading: 1294
Download size: 511 kB / 9,244 kB
Space needed: 2,777 kB / 59.0 GB available
```

```
Continue? [Y/n] Y
Get:1 https://kali.download/kali kali-rolling/main amd64 libtss2-rc0t64 amd64 4.1.3-7 [31.1 kB]
Get:2 https://kali.download/kali kali-rolling/main amd64 systemd-resolved amd64 261.2-1 [322 kB]
Get:3 https://kali.download/kali kali-rolling/main amd64 systemd-tpm amd64 261.2-1 [158 kB]
Fetched 511 kB in 9s (56.2 kB/s)
(Reading database... 426130 files and directories currently installed.)
Preparing to unpack ./libsystemd0_261.2-1_amd64.deb...
Unpacking libsystemd0:amd64 (261.2-1) over (260.1-1)...
Setting up libsystemd0:amd64 (261.2-1)...
(Reading database... 426130 files and directories currently installed.)
Preparing to unpack ./libnss-systemd_261.2-1_amd64.deb...
Unpacking libnss-systemd:amd64 (261.2-1) over (260.1-1)...
Preparing to unpack ./systemd-userdbd_261.2-1_amd64.deb...
Unpacking systemd-userdbd (261.2-1) over (260.1-1)...
Preparing to unpack ./systemd-sysv_261.2-1_amd64.deb...
Unpacking systemd-sysv (261.2-1) over (260.1-1)...
Preparing to unpack ./libpam-systemd_261.2-1_amd64.deb...
Unpacking libpam-systemd:amd64 (261.2-1) over (260.1-1)...
Preparing to unpack ./libsystemd-shared_261.2-1_amd64.deb...
Unpacking libsystemd-shared:amd64 (261.2-1) over (260.1-1)...
Setting up libsystemd-shared:amd64 (261.2-1)...
(Reading database... 426130 files and directories currently installed.)
Preparing to unpack ./libnss-systemd_261.2-1_amd64.deb...
Unpacking libnss-systemd:amd64 (261.2-1) over (260.1-1)...
Preparing to unpack ./systemd-userdbd_261.2-1_amd64.deb...
Unpacking systemd-userdbd (261.2-1) over (260.1-1)...
Preparing to unpack ./systemd-sysv_261.2-1_amd64.deb...
Unpacking systemd-sysv (261.2-1) over (260.1-1)...
Preparing to unpack ./libpam-systemd_261.2-1_amd64.deb...
Unpacking libpam-systemd:amd64 (261.2-1) over (260.1-1)...
Preparing to unpack ./libsystemd-shared_261.2-1_amd64.deb...
Unpacking libsystemd-shared:amd64 (261.2-1) over (260.1-1)...
Setting up libsystemd-shared:amd64 (261.2-1)...
(Reading database... 426063 files and directories currently installed.)
Preparing to unpack ./libtss2-rc0t64_4.1.3-7_amd64.deb...
Unpacking libtss2-rc0t64:amd64 (4.1.3-7)...
Selecting previously unselected package systemd-resolved.
Preparing to unpack ./systemd-resolved_261.2-1_amd64.deb...
Unpacking systemd-resolved (261.2-1)...
Selecting previously unselected package systemd-tpm.
Preparing to unpack ./systemd-tpm_261.2-1_amd64.deb...
```

```
Preparing to unpack ./systemd-tpm_261.2-1_...amd64.deb...
Unpacking systemd-tpm (261.2-1)...
Setting up libtss2-rc0t64:amd64 (4.1.3-7)...
Setting up systemd-tpm (261.2-1)...
Setting up libnss-systemd:amd64 (261.2-1)...
Setting up systemd (261.2-1)...
Installing new version of config file /etc/systemd/networkd.conf...
Installing new version of config file /etc/systemd/system.conf...
Installing new version of config file /etc/systemd/user.conf...
Setting up systemd-timesyncd (261.2-1)...
systemd-time-wait-sync.service is a disabled or a static unit not running, not starting it.
Setting up udev (261.2-1)...
Setting up systemd-resolved (261.2-1)...
Converting /etc/resolv.conf to a symlink to /run/systemd/resolve/stub-resolv.conf...
Creating group 'systemd-resolve' with GID 968.
Creating user 'systemd-resolve' (systemd Resolver) with UID 968 and GID 968.
Created symlink '/etc/systemd/system/dbus-org.freedesktop.resolve1.service' -> '/usr/lib/systemd/system/systemd-resolved.service'.
Created symlink '/etc/systemd/system/sysinit.target.wants/systemd-resolved.service' -> '/usr/lib/systemd/system/systemd-resolved.service'.
Created symlink '/etc/systemd/system/sockets.target.wants/systemd-resolved-varlink.socket' -> '/usr/lib/systemd/system/systemd-resolved-varlink.socket'.
Created symlink '/etc/systemd/system/sockets.target.wants/systemd-resolved-monitor.socket' -> '/usr/lib/systemd/system/systemd-resolved-monitor.socket'.
Setting up systemd-userdbd (261.2-1)...
Setting up systemd-sysv (261.2-1)...
Setting up libpam-systemd:amd64 (261.2-1)...
Processing triggers for initramfs-tools (0.151)...
update-initramfs: Generating /boot/initrd.img-6.19.14+kali-amd64
Processing triggers for libc-bin (2.43-4)...
Processing triggers for man-db (2.13.1-1)...
Processing triggers for dbus (1.16.2-5)...
Processing triggers for shared-mime-info (2.4-5+b3)...
Processing triggers for procs (2:4.0.4-9+b2)...
Processing triggers for kali-menu (2026.2.6)...

(collins@kali) [~/SBT-DF203-Lab7]
$ wget -O evidence/dig_dns.pcap \
  'https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Networking_Forensics/lab_files/dns/dig_dns.pcap'
Prepended http:// to '
--2026-09-21 13:23:09-- http://%20/
Resolving ( )... failed: Name or service not known.
wget: unable to resolve host address '
--2026-09-21 13:23:09-- https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Networking_Forensics/lab_files/dns/dig_dns.pcap
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... failed: Temporary failure in name resolution.
wget: unable to resolve host address 'raw.githubusercontent.com'

(collins@kali) [~/SBT-DF203-Lab7]
$ cp --preserve=timestamps evidence/dig_dns.pcap working/dig_dns_working.pcap

(collins@kali) [~/SBT-DF203-Lab7]
$ sha256sum evidence/dig_dns.pcap working/dig_dns_working.pcap | tee reports/dns_capture_hashes.txt
e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855 evidence/dig_dns.pcap
e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855 working/dig_dns_working.pcap

(collins@kali) [~/SBT-DF203-Lab7]
$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

7. Mini Evidence and Chain-of-Custody Worksheet

Field	Student Entry
Case/lab identifier	SBT-DF203-Lab7-COLLINS KELECHI AMADI
Trainee name	COLLINS AMADI
Date and time started	21-09-2021 01:15:09
Evidence file name(s)	sha256sum evidence/dig_dns.pcap working/dig_dns_working.pcap tee reports/dns_capture_hashes.txt
Source or generation method	COLLINS@KALI
Original SHA-256	sha256sum evidence/dig_dns.pcap working/dig_dns_working.pcap tee reports/dns_capture_hashes.txt

Field	Student Entry
Working-copy SHA-256	3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855 working/dig_dns_working.pcap
Analysis workstation/VM	KALI LINUX COMMAND
Notes on any changes	NILL

8. Part A - Query DNS Records with dig

```
dig example.com A | tee reports/dig_example_A.txt
dig example.com AAAA | tee reports/dig_example_AAAA.txt
dig example.com MX | tee reports/dig_example_MX.txt
dig example.com NS | tee reports/dig_example_NS.txt
dig +short example.com A | tee reports/dig_example_short.txt
```

- Record the resolver shown in the SERVER line.
- Record status/response code, flags, question count, answer count and query time.
- Explain IN, A, AAAA, MX and NS.
- Record at least one TTL and explain that it is a cache lifetime, not a creation date.

ANSWER:

```

Kali Linux - VMware Workstation
File Edit View VM Tabs Help

Library
Type here to search
My Computer
  Owasp bvwa
  Metasploitable 2
  Kali Linux

(collins@kali) ~/SBT-DF203-Lab7
$ dig example.com A | tee reports/dig_example_A.txt

;<<>> DiG 9.20.27-2-Debian <<>> example.com A
;; global options: +cmd
;; Got answer:
;;->HEADER<- opcode: QUERY, status: REFUSED, id: 4893
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;example.com.                IN      A

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Sep 21 13:41:27 WAT 2026
;; MSG SIZE rcvd: 40

(collins@kali) ~/SBT-DF203-Lab7
$ dig example.com AAAA | tee reports/dig_example_AAAA.txt

;<<>> DiG 9.20.27-2-Debian <<>> example.com AAAA
;; global options: +cmd
;; Got answer:
;;->HEADER<- opcode: QUERY, status: REFUSED, id: 52468
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;example.com.                IN      AAAA

```

```
;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Sep 21 13:42:18 WAT 2026
;; MSG SIZE rcvd: 40

(collins@kali)~/SBT-DF203-Lab7
$ dig example.com MX | tee reports/dig_example_MX.txt

;<<>> DiG 9.20.27-2-Debian <<>> example.com MX
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: REFUSED, id: 6989
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;example.com. IN MX

;; Query time: 4 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Sep 21 13:42:35 WAT 2026
;; MSG SIZE rcvd: 40

(collins@kali)~/SBT-DF203-Lab7
$ dig example.com NS | tee reports/dig_example_NS.txt

;<<>> DiG 9.20.27-2-Debian <<>> example.com NS
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: REFUSED, id: 59171
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;example.com. IN NS

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Sep 21 13:42:50 WAT 2026
;; MSG SIZE rcvd: 40

(collins@kali)~/SBT-DF203-Lab7
$ dig +short example.com A | tee reports/dig_example_short.txt
```

9. Part B - Capture a Fresh DNS Query

IFACE=eth0

sudo tshark -i "\$IFACE" -f 'port 53' -a duration:25 -w evidence/fresh_dig_dns.pcapng &

sleep 3

dig +noedns example.com A >/dev/null

wait

sha256sum evidence/fresh_dig_dns.pcapng | tee reports/fresh_dns_sha256.txt

ANSWER:

```
Kali Linux - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Owasp bwwa
Metasploitable 2
Kali Linux

(collins@kali)~/SBT-DF203-Lab7
$ IFACE=eth0

(collins@kali)~/SBT-DF203-Lab7
$ sudo tshark -i "$IFACE" -f 'port 53' -a duration:25 -w evidence/fresh_dig_dns.pcapng &
[1] 87656

(collins@kali)~/SBT-DF203-Lab7
$ Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'

[1] + done sudo tshark -i "$IFACE" -f 'port 53' -a duration:25 -w
(collins@kali)~/SBT-DF203-Lab7
$ sleep 3

(collins@kali)~/SBT-DF203-Lab7
$ dig +noedns example.com A >/dev/null

(collins@kali)~/SBT-DF203-Lab7
$ wait

(collins@kali)~/SBT-DF203-Lab7
$ sha256sum evidence/fresh_dig_dns.pcapng | tee reports/fresh_dns_sha256.txt
c65b5045297eb3d41311fcfe8d7be0a85b177e3ab065b42a3bd40ab247effe76 evidence/fresh_dig_dns.pcapng

(collins@kali)~/SBT-DF203-Lab7
$
```

Interface Note

On systems using a local stub resolver, packets may appear on lo and then on a physical interface. Use tshark -D to list interfaces and choose the one that shows the actual query.

10. Part C - Extract DNS Query and Response Fields

PCAP=evidence/fresh_dig_dns.pcapng

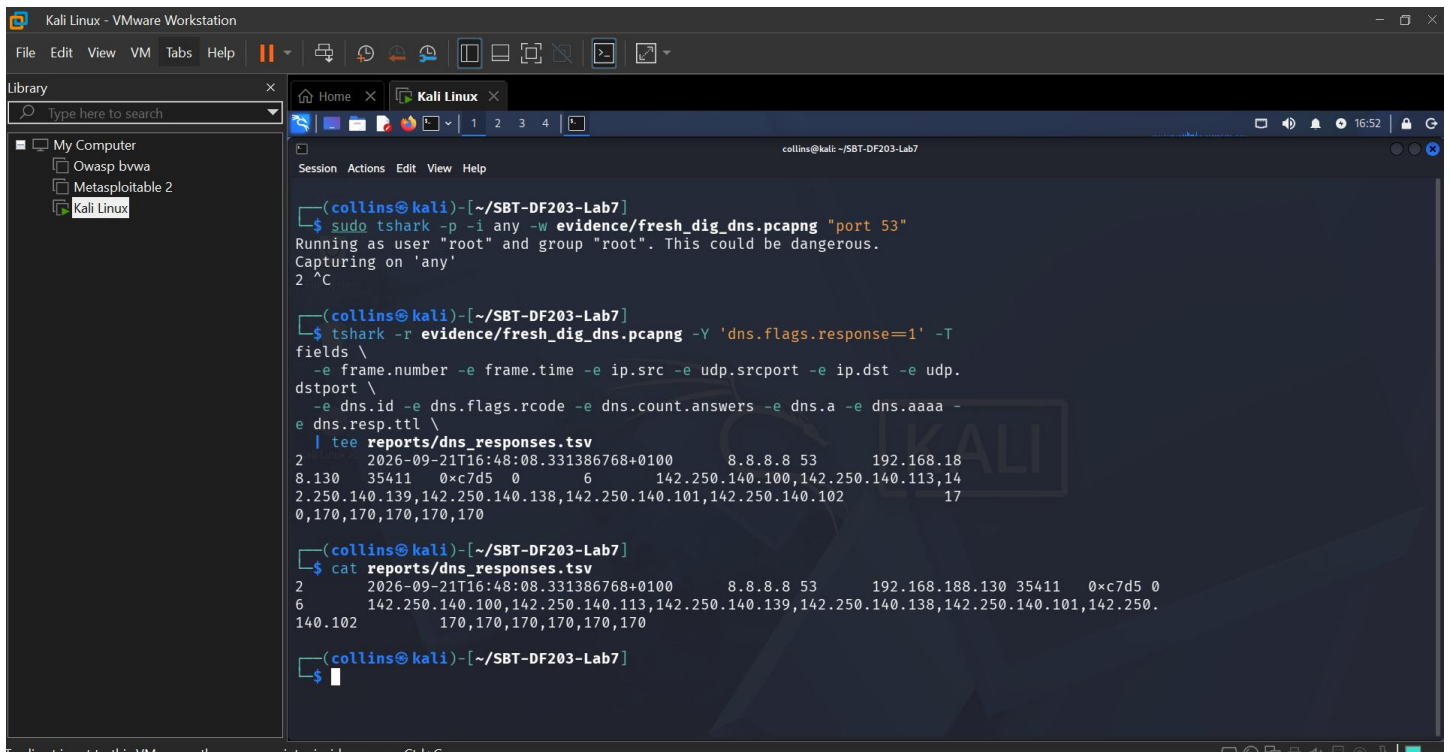
Query

```
tshark -r "$PCAP" -Y 'dns.flags.response==0' -T fields \
-e frame.number -e frame.time -e ip.src -e udp.srcport -e ip.dst -e udp.dstport \
-e dns.id -e dns.qry.name -e dns.qry.type \
| tee reports/dns_queries.tsv
```

Response

```
tshark -r "$PCAP" -Y 'dns.flags.response==1' -T fields \
-e frame.number -e frame.time -e ip.src -e udp.srcport -e ip.dst -e udp.dstport \
-e dns.id -e dns.flags.rcode -e dns.count.answers -e dns.a -e dns.aaaa -e dns.resp.ttl \
| tee reports/dns_responses.tsv
```

ANSWER:



```
Kali Linux - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Owasp bwba
Metasploitable 2
Kali Linux

(collins@kali) - [~/SBT-DF203-Lab7]
$ sudo tshark -p -i any -w evidence/fresh_dig_dns.pcapng "port 53"
Running as user "root" and group "root". This could be dangerous.
Capturing on 'any'
2 ^C

(collins@kali) - [~/SBT-DF203-Lab7]
$ tshark -r evidence/fresh_dig_dns.pcapng -Y 'dns.flags.response==1' -T
fields \
-e frame.number -e frame.time -e ip.src -e udp.srcport -e ip.dst -e udp.
dstport \
-e dns.id -e dns.flags.rcode -e dns.count.answers -e dns.a -e dns.aaaa -
e dns.resp.ttl \
| tee reports/dns_responses.tsv
2      2026-09-21T16:48:08.331386768+0100      8.8.8.8 53      192.168.18
8.130  35411      0xc7d5 0      6      142.250.140.100,142.250.140.113,14
2.250.140.139,142.250.140.138,142.250.140.101,142.250.140.102      17
0,170,170,170,170,170

(collins@kali) - [~/SBT-DF203-Lab7]
$ cat reports/dns_responses.tsv
2      2026-09-21T16:48:08.331386768+0100      8.8.8.8 53      192.168.188.130 35411      0xc7d5 0
6      142.250.140.100,142.250.140.113,142.250.140.139,142.250.140.138,142.250.140.101,142.250.
140.102      170,170,170,170,170,170

(collins@kali) - [~/SBT-DF203-Lab7]
$
```

11. Part D - Match Queries to Responses

Transaction ID	Query Name	Type	Client/Resolver	Response Code	Answer(s)	TTL	Time Delta
0xc7d5	google.com	A	35411/53	291248977	False		2026-09-21T16:48:08
0xc7d5	google.com	NOERROR	35411/53	331386768	True		2026-09-21T16:48:08
	google.com						

- Match query and response using transaction ID plus the endpoint tuple.
- Confirm that the response source is the queried resolver.
- Calculate response time from frame timestamps.
- Document CNAME chains and multiple A/AAAA answers where present.

12. Part E - Analyze DNS Generated by a Browser

A browser may request several domains for the page, images, fonts, telemetry or cached services. Capture a visit to a non-sensitive instructor-approved site and inventory all query names.

```
sudo tshark -i IFACE -f 'port 53' -a duration:40 -w evidence/browser_dns.pcapng &
sleep 3
# Open the approved site in a private browser window, then wait.
wait
```

```
tshark -r evidence/browser_dns.pcapng -Y 'dns.flags.response==0' -T fields -e dns.qry.name -e dns.qry.type \
| sort | uniq -c | sort -nr | tee reports/browser_dns_inventory.txt
```

ANSWER:

```

(collins@kali)~/SBT-DF203-Lab7
$ sudo tshark -i eth0 -f 'port 53' -a duration:40 -w evidence/browser_dns.pcapng & sleep 3
[1] 236915
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'

(collins@kali)~/SBT-DF203-Lab7$
30
[1] + done sudo tshark -i eth0 -f 'port 53' -a duration:40 -w evidence/browser_dns.pcapng
(collins@kali)~/SBT-DF203-Lab7$
$ sudo tshark -r evidence/browser_dns.pcapng \
-y 'dns.flags.response==0' \
-T fields \
-e frame.number \
-e frame.time \
-e ip.src \
-e ip.dst \
-e udp.srcport \
-e udp.dstport \
-e dns.id \
-e dns.qry.name \
-e dns.qry.type \
| tee ~/reports/browser_dns_inventory.txt
Running as user "root" and group "root". This could be dangerous.
1 2026-09-21T22:06:05.596205322+0100 192.168.188.130 8.8.8.8 53158 53 0xf4ef example.org 1
2 2026-09-21T22:06:05.596476540+0100 192.168.188.130 8.8.8.8 49236 53 0x8cd5 cloudflare-dns.com 65
3 2026-09-21T22:06:05.596592634+0100 192.168.188.130 8.8.8.8 34231 53 0x50aa example.org 1
4 2026-09-21T22:06:05.596697746+0100 192.168.188.130 8.8.8.8 34231 53 0xb6ac example.org 28

```

```
5      2026-09-21T22:06:05.596843383+0100      192.168.188.130 8.8.8.8 46234 53      0x90ea ipv4only.arpa 1
6      2026-09-21T22:06:05.596965059+0100      192.168.188.130 8.8.8.8 46234 53      0x86f7 ipv4only.arpa 28
7      2026-09-21T22:06:05.598435365+0100      192.168.188.130 8.8.8.8 40445 53      0x937f detectportal.firefox.com 1
8      2026-09-21T22:06:05.600022321+0100      192.168.188.130 8.8.8.8 52815 53      0xc01e detectportal.firefox.com 1
9      2026-09-21T22:06:05.600138477+0100      192.168.188.130 8.8.8.8 52815 53      0x7a1c detectportal.firefox.com 2
8
19     2026-09-21T22:06:08.093241612+0100      192.168.188.130 8.8.8.8 33630 53      0x3691 example.com 65
20     2026-09-21T22:06:08.093429285+0100      192.168.188.130 8.8.8.8 37740 53      0x3789 example.com 1
21     2026-09-21T22:06:08.093434680+0100      192.168.188.130 8.8.8.8 37740 53      0x6e8a example.com 28
22     2026-09-21T22:06:08.126479567+0100      192.168.188.130 8.8.8.8 41650 53      0xb11f example.com 1
27     2026-09-21T22:06:19.869339795+0100      192.168.188.130 8.8.8.8 40699 53      0x2428 push.services.mozilla.com 1
28     2026-09-21T22:06:19.869661368+0100      192.168.188.130 8.8.8.8 40699 53      0x33a6 push.services.mozilla.com 2
8

(collins@kali) - [~/SBT-DF203-Lab7]
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

13. Part F - Correlate DNS with Subsequent Connections

Extract DNS A answers

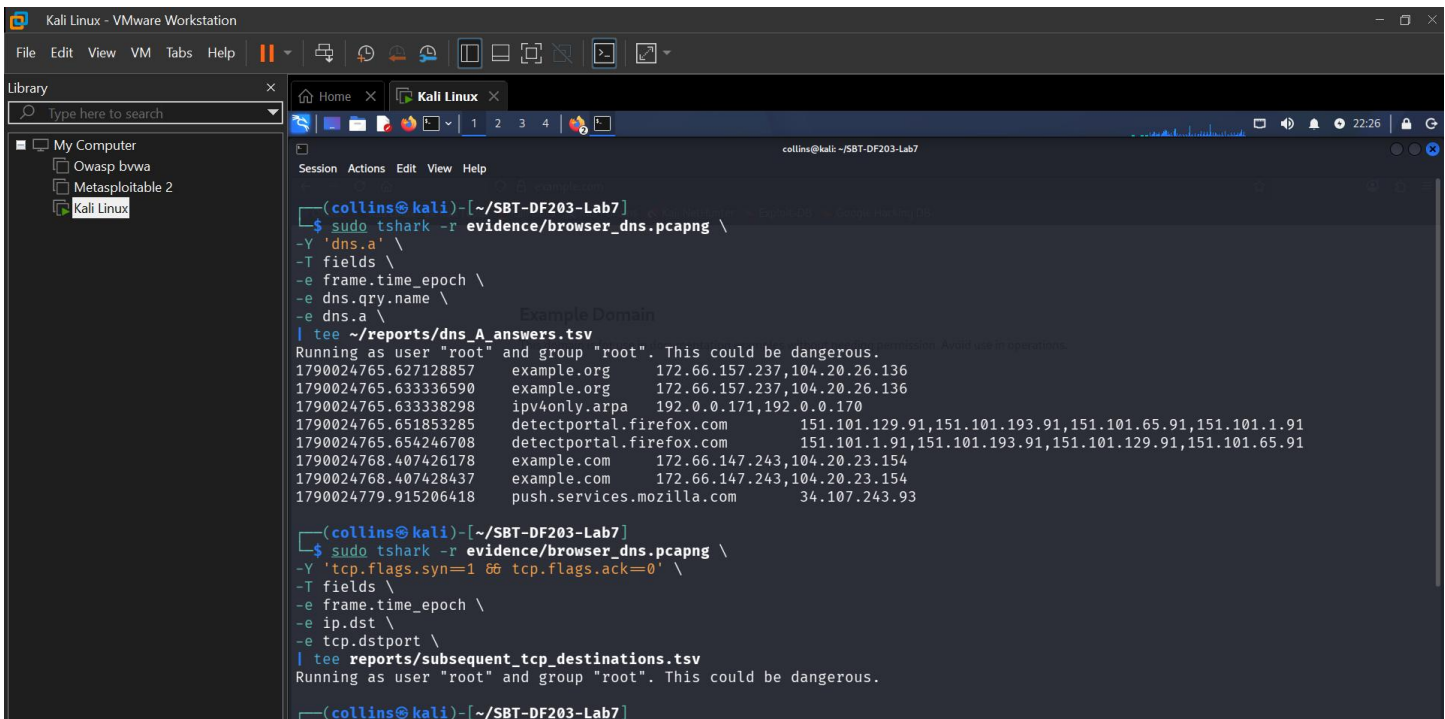
```
tshark -r evidence/browser_dns.pcapng -Y 'dns.a' -T fields -e frame.time_epoch -e dns.qry.name -e dns.a \
| tee reports/dns_A_answers.tsv
```

Extract later TCP destinations

```
tshark -r evidence/browser_dns.pcapng -Y 'tcp.flags.syn==1 && tcp.flags.ack==0' -T fields \
-e frame.time_epoch -e ip.dst -e tcp.dstport \
| tee reports/subsequent_tcp_destinations.tsv
```

Select one DNS answer and show whether the client later initiates a connection to that IP. Explain any mismatch caused by multiple answers, proxies, CDNs, IPv6, caching or capture duration.

ANSWER:



```
Kali Linux - VMware Workstation
File Edit View VM Tabs Help

Library
Type here to search
My Computer
  Owasp bwba
  Metasploitable 2
  Kali Linux

(collins@kali) - [~/SBT-DF203-Lab7]
$ sudo tshark -r evidence/browser_dns.pcapng \
-Y 'dns.a' \
-T fields \
-e frame.time_epoch \
-e dns.qry.name \
-e dns.a \
| tee ~/reports/dns_A_answers.tsv
Running as user "root" and group "root". This could be dangerous.
1790024765.627128857 example.org 172.66.157.237,104.20.26.136
1790024765.633336590 example.org 172.66.157.237,104.20.26.136
1790024765.633338298 ipv4only.arpa 192.0.0.171,192.0.0.170
1790024765.651853285 detectportal.firefox.com 151.101.129.91,151.101.193.91,151.101.65.91,151.101.1.91
1790024765.654246708 detectportal.firefox.com 151.101.1.91,151.101.193.91,151.101.129.91,151.101.65.91
1790024768.407426178 example.com 172.66.147.243,104.20.23.154
1790024768.407428437 example.com 172.66.147.243,104.20.23.154
1790024779.915206418 push.services.mozilla.com 34.107.243.93

(collins@kali) - [~/SBT-DF203-Lab7]
$ sudo tshark -r evidence/browser_dns.pcapng \
-Y 'tcp.flags.syn==1 && tcp.flags.ack==0' \
-T fields \
-e frame.time_epoch \
-e ip.dst \
-e tcp.dstport \
| tee reports/subsequent_tcp_destinations.tsv
Running as user "root" and group "root". This could be dangerous.

(collins@kali) - [~/SBT-DF203-Lab7]
```

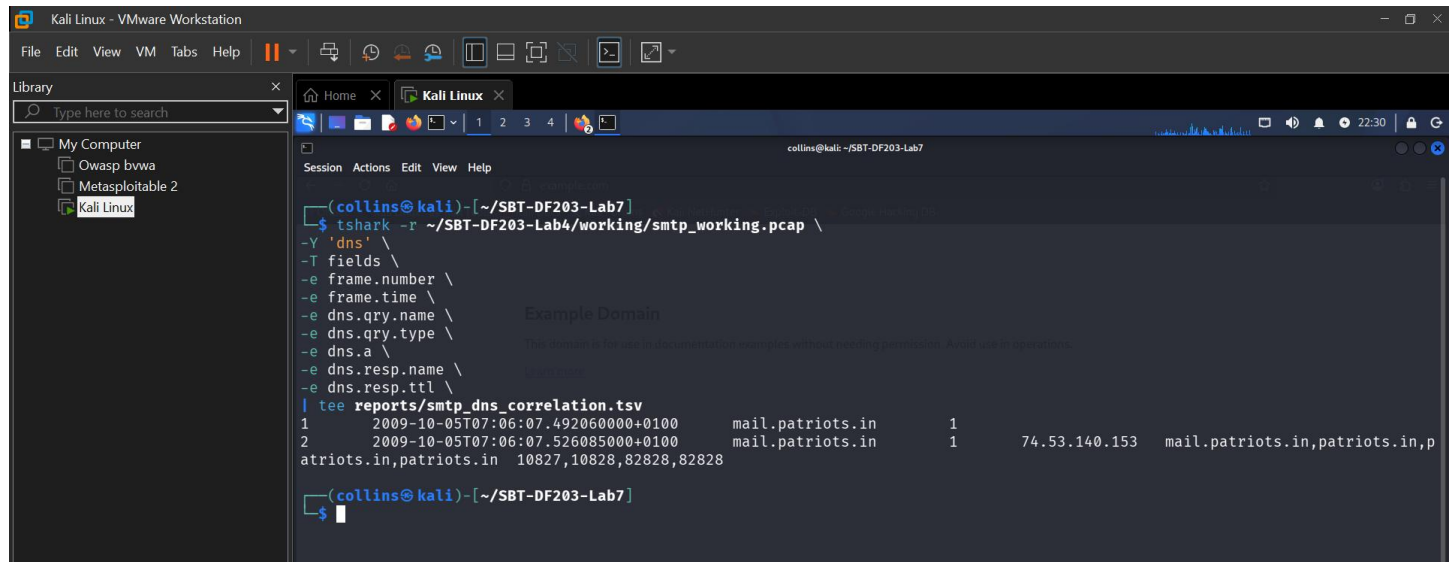
14. Part G - DNS Analysis Within SMTP Evidence

If smtp.pcap from Lab 4 is available, filter dns and identify the query used to locate a mail server. Record the queried name, record type, resolver, answer and TTL.

Optional, adjust path to your Lab 4 capture

```
tshark -r ../SBT-DF203-Lab4/working/smtp_working.pcap -Y 'dns' -T fields \
-e frame.number -e frame.time -e dns.qry.name -e dns.qry.type -e dns.a -e dns.resp.name -e dns.resp.ttl \
| tee reports/smtp_dns_correlation.tsv
```

ANSWER:



15. Required Findings Worksheet

Field	Finding
Configured resolver IP	192.168.188.130
Client source port	37740
Resolver destination port	53
Transaction ID	0x3789
Query name and type	example.com
Response code	A
Answer IP(s)	172.66.147.243, 104.20.23.154,
TTL	093429285

Field	Finding
Query-response time delta	2026-09-21T22:06:08
Subsequent connection correlation	"Yes, a later outbound TCP connection on port 53 was observed from host 192.168.188.130 to the returned IP at 06:08."
Normal variations observed	

Required Screenshots Checklist

- ☐ Resolver configuration.
- ☐ dig A/AAAA/MX/NS outputs.
- ☐ Fresh DNS capture hash.
- ☐ DNS query fields.
- ☐ Matching DNS response fields.
- ☐ Transaction ID correlation.
- ☐ Answer and TTL.
- ☐ Browser query inventory.
- ☐ DNS answer to TCP destination correlation.
- ☐ Optional SMTP DNS evidence.

Student Report Template

1. Executive summary.
2. Resolver and environment.
3. Evidence integrity.
4. dig results.
5. Packet-level query/response analysis.
6. Transaction matching and timing.
7. Browser DNS inventory.
8. Connection correlation.
9. Limitations and normal variations.

10. Appendix.

Submission Requirements

- One PDF report named SBT-DF203-Lab7_RegNo_FullName.pdf.
- The original or generated PCAP/PCAPNG evidence file and its SHA-256 hash text file.
- A reports folder containing exported CSV/TXT command output and any recovered objects.
- Screenshots must be readable, numbered and referenced in the report narrative.
- Compress the report and evidence outputs into one ZIP named with your registration number and lab number.
- Do not submit passwords or decoded credentials in public discussion channels; include them only in the protected assessment submission where required.

Marking Rubric (100 Marks)

Assessment Area	Marks	What the Instructor Will Check
Preparation and resolver identification	10	Correct resolver and evidence hashes.
dig record analysis	15	A, AAAA, MX, NS and TTL interpretation.
Packet-field extraction	25	Correct IDs, names, types, endpoints and response codes.
Query-response correlation	20	Accurate matching and timing.
Browser/connection correlation	20	Inventory and defensible IP connection linkage.
Report quality	10	Clear screenshots, tables and conclusions.

Instructor Quick Answer Guide

Checkpoint	Expected Observation
Standard transport	Usually UDP destination port 53; TCP may be used for large responses or zone transfer.
Query flag	dns.flags.response==0.
Response flag	dns.flags.response==1.
A record	IPv4 address.
AAAA record	IPv6 address.

Checkpoint	Expected Observation
TTL	How long a resolver may cache the record.

Command Reference Summary

Command	Purpose
dig NAME TYPE	Query a DNS record.
dns.qry.name	Queried domain name.
dns.qry.type	Requested record type.
dns.id	DNS transaction identifier.
dns.flags.rcode	Response code.
dns.resp.ttl	Answer TTL.

Academic Integrity and Authorization Statement

Student Declaration

I confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.